

Raster Data

Objectives:	Understand what raster data is and how it can be used in a GIS.
Keywords:	Raster, Pixel, Remote Sensing, Satellite, Image, Georeferenced

Overview

- In the previous topics we have taken a closer look at **vector data**.
- While **vector features use geometry** (points, polylines and polygons) to represent the real world, raster data takes a different approach.
- Raster's are **made up of a matrix of pixels** (also called **cells**), each containing a value that represents the conditions for the area covered by that cell (see [Fig.1.](#)).
- **In this topic** we are going to take a closer look at raster data, when it is useful and when it makes more sense to use vector data.

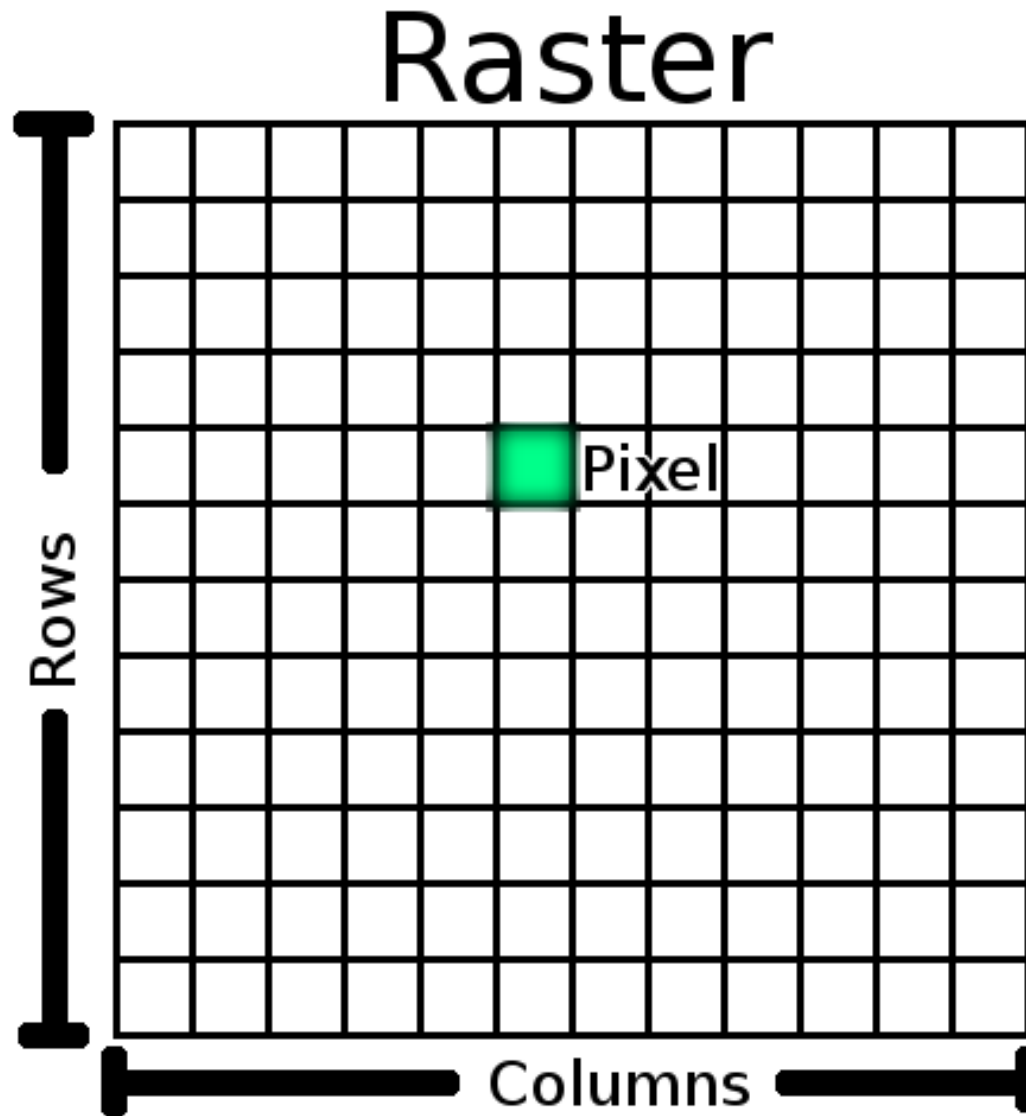


Fig. 1. A raster dataset is composed of rows (running across) and columns (running down) of pixels (also know as cells). Each pixel represents a geographical region, and the value in that *pixel* represents some *characteristic of that region*.

Raster data in detail

Raster data is used in a GIS application when we want

- To display information that is continuous across an area and
- cannot easily be divided into vector features.



Example

- *Some features on a landscape are easy to represent as points, polylines and polygons (e.g. trees, roads, houses).*
- *In other cases it can be difficult.*
- *For example how would you represent the grasslands?*
- *As polygons?*
- *What about the variations in color you can see in the grass?*
- ***When you are trying to represent large areas with continuously changing values, raster data can be a better choice.***

So, Why we Use Rater Data??

- It would be easy enough to make a single polygon around each grassland area, but **a lot of the information about the grassland would be lost** in the process of simplifying the features to a single polygon.
- This is because when you give a vector feature **attribute values**, they apply to the **whole feature**,
- so **vectors aren't very good at representing** features that are not **homogeneous (entirely the same)** all over.
- **Another approach you could take is to**
- The problem with that approach is that it will take a huge amount of work in order to create a good vector dataset.

- Using raster data is a solution to these problems.
- Many people use raster data as a **backdrop** to be used **behind vector layers** in order **to provide more meaning to the vector information**.
- The human eye is very good at interpreting images and so using an image behind vector layers, results in maps with a lot more meaning.

- Raster data is not only good for images that depict the real world surface (e.g. satellite images and aerial photographs), they are also good for representing more abstract ideas.

For example,

- raster's can be used to show rainfall trends over an area, or to depict the fire risk on a landscape.
- In these kinds of applications, each cell in the raster represents a different value e.g. risk of fire on a scale of one to ten.

An example that shows the difference between an image obtained from a satellite and one that shows calculated values can be seen in Fig.3.

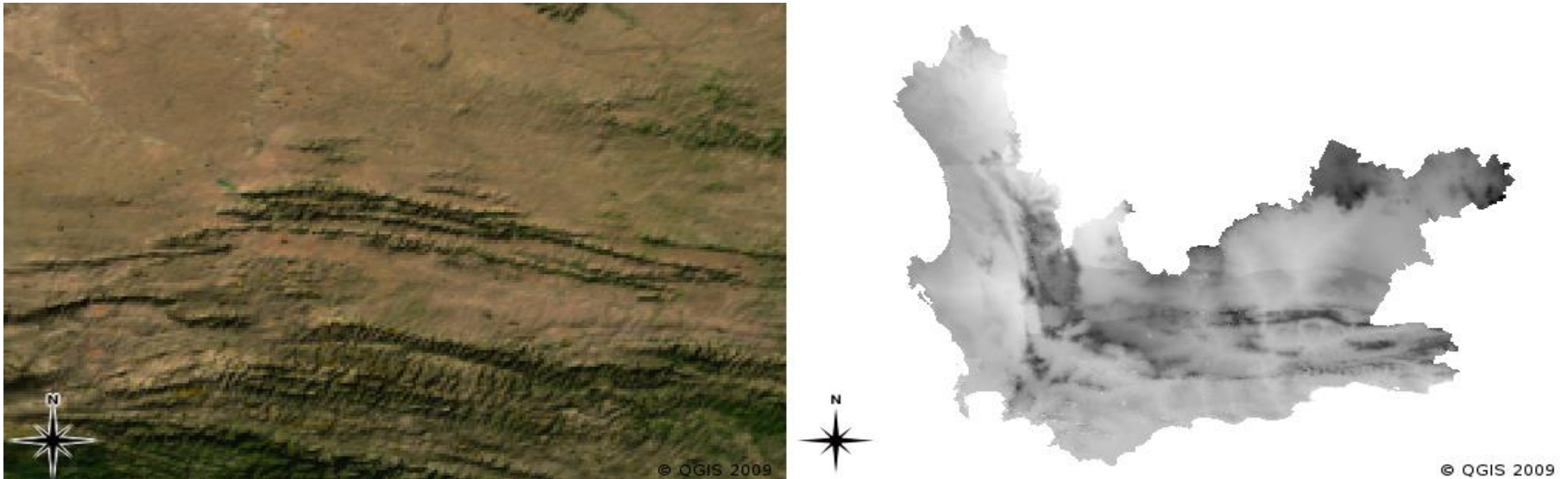


Fig. 3.

- True color raster images (*left*) are useful as they provide a lot of detail that is hard to capture as vector features but easy to see when looking at the raster image.
- Raster data can also be non-photographic data such as the *raster layer shown on the right* which shows the *calculated average minimum temperature* in the Western Cape for the month of March.

Georeferencing

- Georeferencing is the **process of defining exactly where on the earth's surface** an image or raster **dataset was created**.
- This positional information is stored with the digital version of the aerial photo.
- When the GIS application opens the photo, it uses the positional information to ensure that the photo appears in the correct place on the map.

- Normally this positional information consists of a coordinate for the top left pixel in the image,
- The **size of each pixel in the X direction**, the **size of each pixel in the Y direction**, and the amount (if any) by which the image is rotated.
- With these few pieces of information, the GIS application can ensure that raster data are displayed in the correct place.
- The georeferencing information for a raster is often provided in a **small text file accompanying the raster**.

Sources of raster data

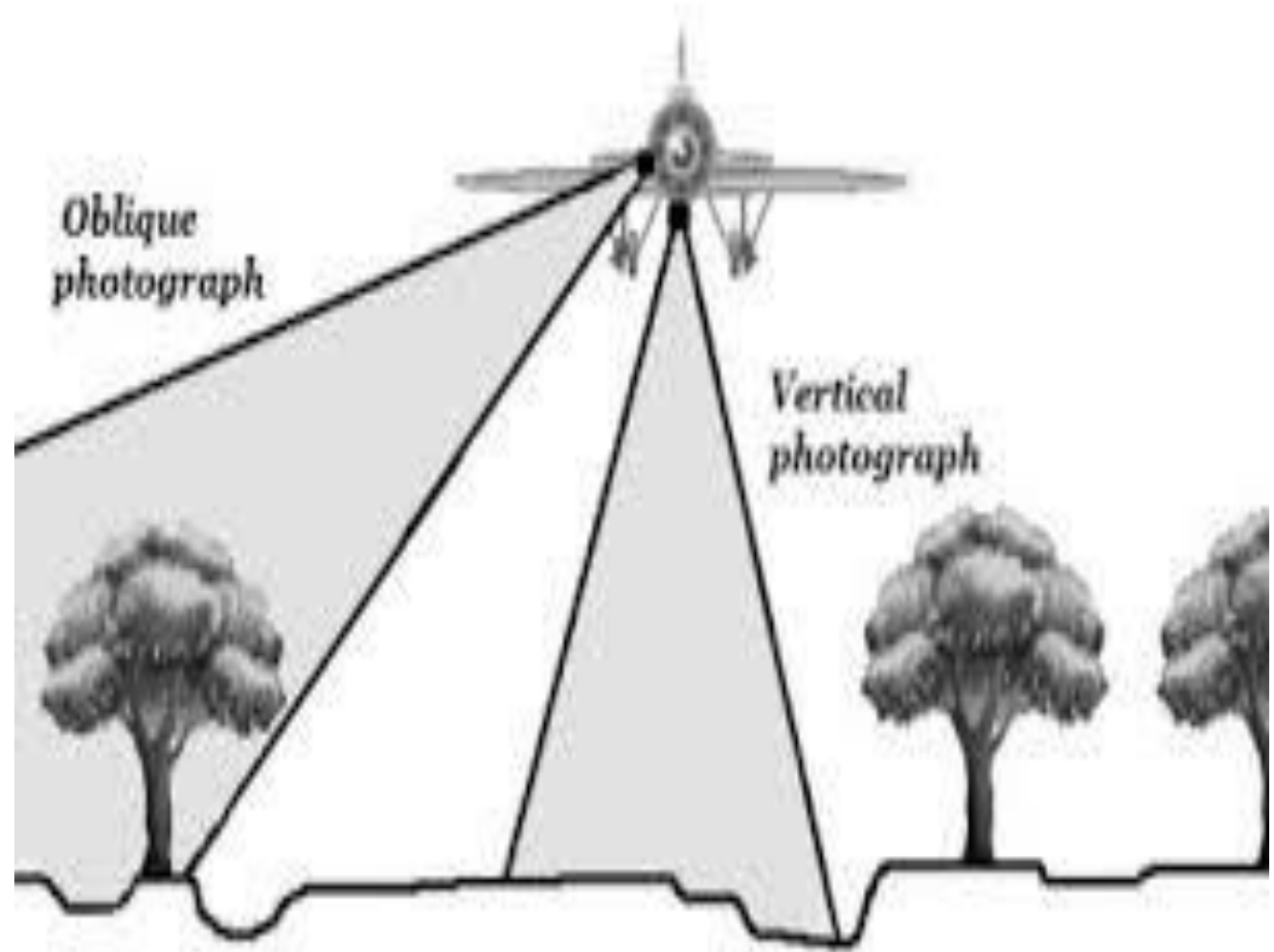
- Raster data can be obtained in a number of ways.
- Two of the most common ways are

➤ **Aerial photography and**

➤ **Satellite imagery.**

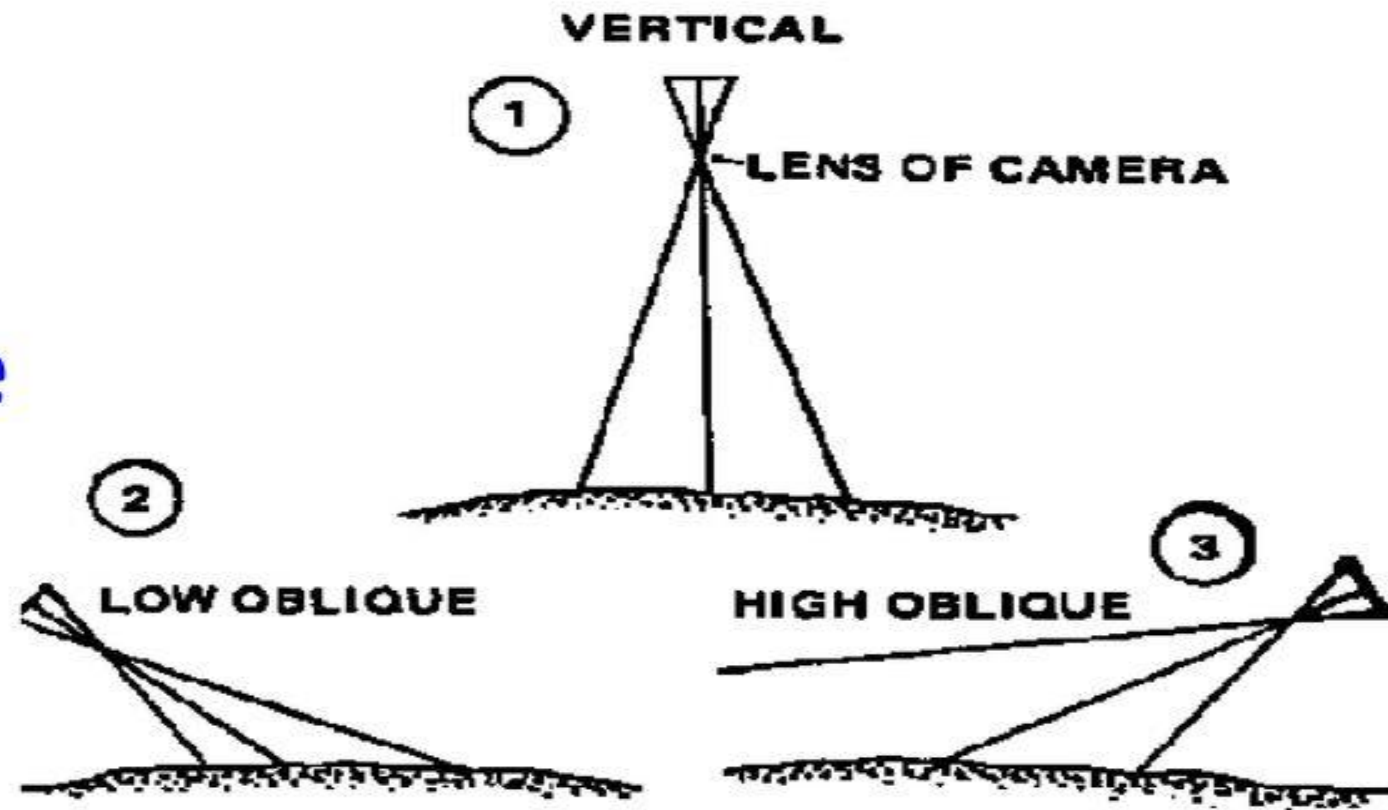
Aerial photography

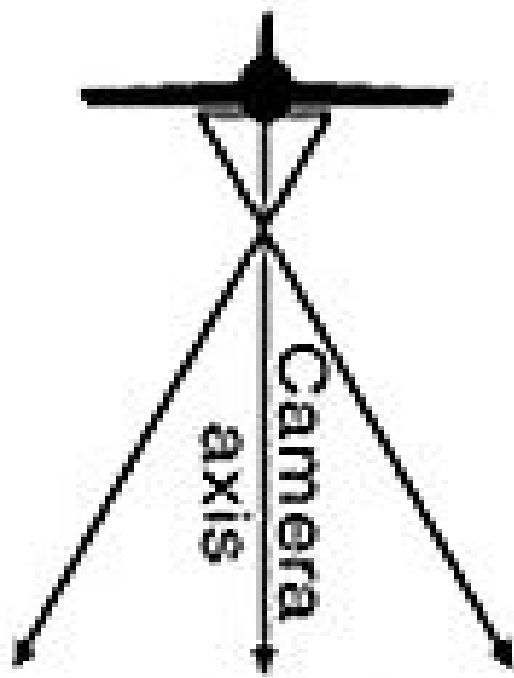
- In aerial photography, an aeroplane flies over an area with a camera mounted underneath it.
- The photographs are then imported into a computer and georeferenced.



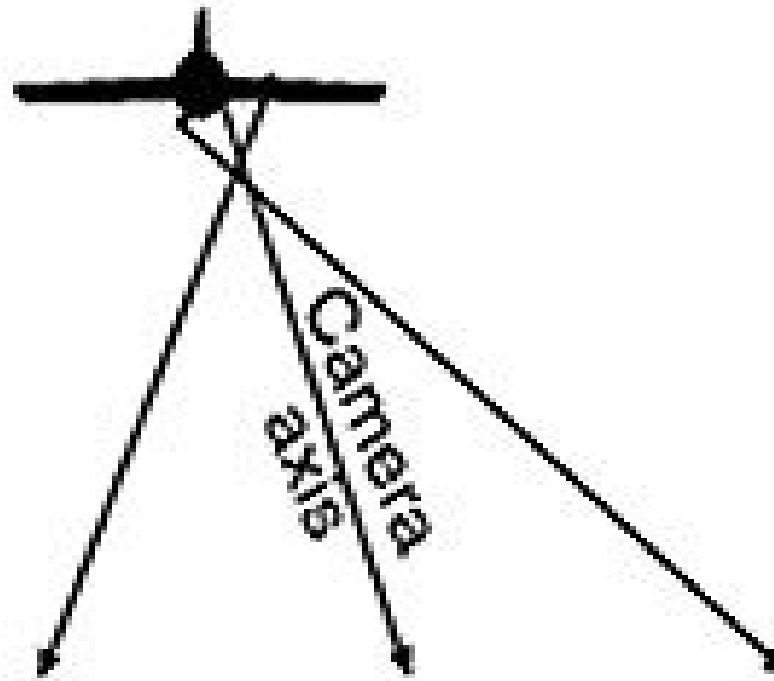
A) TYPES OF AERIAL PHOTOGRAPHS ON ANGLES

1. Vertical
2. Low oblique
3. High oblique

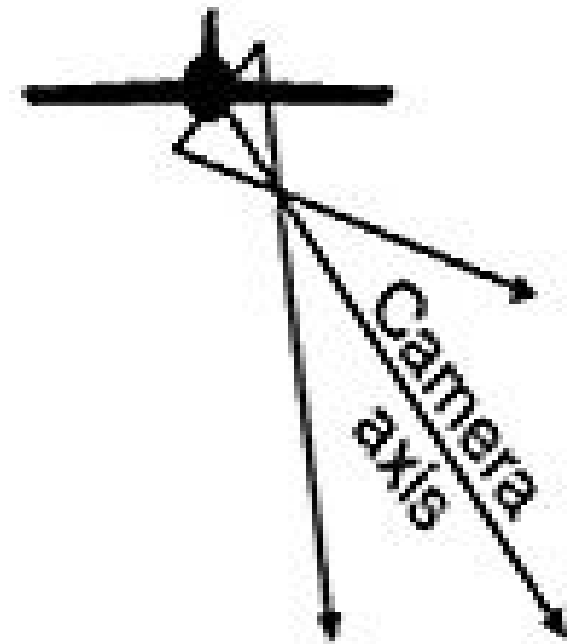




Vertical



Low oblique



High oblique





Satellite imagery

- Satellite imagery is created when satellites orbiting the earth point special digital cameras towards the earth and then take an image of the area on earth they are passing over.
- Once the image has been taken it is sent back to earth using radio signals to special receiving stations such as the one shown in Fig.7.



EPS Contributes to the Global Operational Satellite Observation System



Remote sensing

- The process of capturing raster data from an Aeroplane or satellite is called **remote sensing**.

In other cases, raster data can be computed.

For example

- an **insurance company** may take **police crime incident** reports and create a country wide raster **map showing how high the incidence of crime** is likely to be in each area.
- **Meteorologists** (people who study weather patterns) might generate a province level raster showing average temperature, rainfall and wind direction using data collected from weather stations

Next: Raster Resolution.....

Raster Resolution

- **Spatial Resolution**
- **Spectral resolution**

Spatial Resolution

- Every raster layer in a GIS has pixels (cells) of a fixed size that determine its spatial resolution.
- This becomes apparent when you look at an image at a small scale (see Fig. 8) and then zoom in to a large scale (see Fig. 9).

Fig. 8 *This satellite image looks good when using a small scale...*



Fig. 9 ...but when viewed at a large scale you can see the individual pixels that the image is composed of.



Factors:

Which determine the spatial resolution of an image.

- For **remote sensing** data, spatial resolution is usually determined by the capabilities of the sensor used to take an image.
- For example **SPOT5** satellites can take images where each pixel is 10 m x 10 m.
- Other satellites, for example **MODIS** take images only at 500 m x 500 m per pixel.
- In **aerial photography**, pixel sizes of 50 cm x 50 cm are not uncommon.

spatial resolution

- Images with a pixel size covering a small area are called '**high resolution**' images because it is possible to make out a high degree of detail in the image.
- Images with a pixel size covering a large area are called '**low resolution**' images because the amount of detail the images show is low.

Example...

- In raster data that is computed by spatial analysis (such as the rainfall map we mentioned earlier), the spatial **density of information** used to create the raster will usually **determine the spatial resolution**.
- **For example** if you want to create a high resolution average rainfall map, you would ideally need many weather stations in close proximity to each other.

Where we use Low spatial resolution data?

- Sometimes using a low spatial resolution is useful when you want to work with a **large area** and are **not interested in looking at any one area** in a lot of detail.
- The cloud maps you see on the weather report, are an example of this — it's useful to see the clouds across the whole country.
- **Zooming in to one particular cloud in high resolution will not tell you very much about the upcoming weather!**

Spectral resolution

- If you take a color photograph with a digital camera or camera on a cellphone, the camera uses electronic sensors to detect red, green and blue light.
- When the picture is displayed on a screen or printed out, the red, green and blue (RGB) information is combined to show you an image that your eyes can interpret.
- While the information is still in digital format though, this RGB information is stored in separate color **bands**.

- our eyes can only see RGB wavelengths,
- the electronic sensors in cameras are able to detect wavelengths that our eyes cannot.
- Of course in a hand held camera it probably doesn't make sense to record information from the **non-visible** parts of the spectrum since most people just want to look at pictures of their or what have you.
- Raster images that include data for **non-visible** parts of the **light spectrum** are often referred to as **multi-spectral images**.
- In **GIS** recording the non-visible parts of the spectrum can be very useful. **For example,**
Measuring **infra-red light** can be useful in identifying **water bodies**.

spectral resolution

- Because having images containing multiple bands of light is so useful in GIS, raster data are often provided as multi-band images.
- Each band in the image is like a separate layer.
- The GIS will combine three of the bands and show them as red, green and blue so that the human eye can see them.
- The number of bands in a raster image is referred to as its **spectral resolution**.

Image types.

- If an image consists of only one band, it is often called a **grayscale** image.
- With grayscale images, you can apply false coloring to make the differences in values in the pixels more obvious.
- Images with false coloring applied are often referred to as **pseudocolor images**.

Raster/vector conversion

- **Raster to vector conversion**
- **Vector to Raster conversion**

Raster to vector conversion

- In our discussion of vector data, we explained that often raster data are used as a **backdrop layer**, which is then used **as a base** from which vector features can be digitized.
- **Another approach** is to use advanced computer programs to **automatically extract vector features** from images.
- Some features such as roads show in an image as a sudden change of color from neighboring pixels.
- The computer program looks for such color changes and creates vector features as a result. This kind of functionality is normally only available in very specialized (and often expensive) GIS software.

Vector to raster conversion

- Sometimes it is useful to convert vector data into raster data.
- One **side effect** of this is that attribute data (that is attributes associated with the original vector data) will be lost when the conversion takes place.
- Having vectors converted to raster format can be useful though when you want **to give GIS data to non GIS users**.
- With the simpler raster formats, the person you give the raster image to can simply view it as an image on their computer without needing any special GIS software.

Raster analysis

- There are a great many analytical tools that can be run on raster data which cannot be used with vector data.

For example,

- Raster's can be used to model water flow over the land surface.
- This information can be used to calculate where watersheds and stream networks exist, based on the terrain.

Agriculture and forestry Management

- Raster data are also often used in agriculture and forestry to manage crop production.

For example

- with a satellite image of a farmer's lands, you can identify areas where the plants are growing poorly and then use that information to apply more fertilizer on the affected areas only.
- Foresters use raster data to estimate how much timber can be harvested from an area.

Disaster management

- Raster data is also very important for disaster management.
- Analysis of **Digital Elevation Models** (a kind of raster where each pixel contains the height above sea level) can then be used to identify areas that are likely to be flooded.
- This can then be used to target rescue and relief efforts to areas where it is needed the most.

What have we learned?

- Let's wrap up what we covered in this worksheet:
- Raster data are a grid of regularly sized **pixels**.
- Raster data are good for showing **continually varying information**.
- The size of pixels in a raster determines its **spatial resolution**.
- Raster images can contain one or more **bands**, each covering the same spatial area, but containing different information.
- When raster data contains bands from different parts of the electromagnetic spectrum, they are called **multi-spectral images**.
- Three of the bands of a multi-spectral image can be shown in the colours Red, Green and Blue so that we can see them.
- Images with a single band are called grayscale images.
- Single band, grayscale images can be shown in pseudocolor by the GIS.
- Raster images can consume a large amount of storage space.

Now you try!

Here are some ideas for you to try :

- Discuss with your fellows which situations you would use raster data and in which you would use vector data.
- create a raster map of your school by using A4 transparency sheets with grid lines drawn on them. Overlay the transparencies onto a toposheet or aerial photograph of your school. Now let each member or group color in cells that represent a certain type of feature e.g. building, playground, sports field, trees, footpaths etc.
- When you are all finished, overlay all the sheets together and see if it makes a good raster map representation of your school.
- Which types of features worked well when represented as rasters? How did your choice in cell size affect your ability to represent different feature types?

Something to think about

- If you don't have a computer available, you can understand raster data using pen and paper.
- Draw a grid of squares onto a sheet of paper to represent your soccer field.
- Fill the grid in with numbers representing values for grass cover on your soccer field.
- If a patch is bare give the cell a value of 0.
- If the patch is mixed bare and covered, give it a value of 1.
- If an area is completely covered with grass, give it a value of 2.
- Now use pencil crayons to color the cells based on their values.
- Color cells with **value 2** dark green.
- **Value 1** should get colored light green, and **value 0** colored in brown.
- When you finish, you should have a raster map of your **soccer field!**

Further reading

Book:

- Chang, Kang-Tsung (2006). Introduction to Geographic Information Systems. 3rd Edition. McGraw Hill. ISBN: 0070658986
- DeMers, Michael N. (2005). Fundamentals of Geographic Information Systems. 3rd Edition. Wiley. ISBN: 9814126195

Website: https://en.wikipedia.org/wiki/GIS_file_formats#Raster

- The QGIS/ArcGis User Guide also has more detailed information on working with raster data..

What's next?

- In the section that follows we will take a closer look at **topology** to see how the relationship between vector features can be used to ensure the best data quality.